

Auto-Graded Homework that Invites Students to Think Broadly

Mark Eichenlaub · Art of Problem Solving

The default ask: "find the number"

Homework that grades itself usually presents a scenario and asks for one quantity. Such problems build mastery and tenacity — but a steady diet of them trains a narrow, procedural picture of what doing physics *is*: pick the equation, plug in, solve.

Find the tension in the rope.

Find the time to reach the ground.

Find the final speed.

Find the distance traveled.

Find the ...

The grader isn't the limit — the question format is

An auto-grader can check any answer with a well-defined key — not just numbers.

Change *what you ask for*, and the same instant-feedback machinery invites the patterns of thought we prize in rich human-feedback settings: designing experiments, choosing models, weighing evidence, critiquing calculations, diagnosing errors.

If the answer is ___, the thinking is ___

a number ± tolerance	execute a plan
items sorted into bins	apply a criterion consistently
two columns connected	map structure onto structure
all-that-apply choices	judge each claim on its own
a click on the image	commit to a spatial prediction
a ranking	compare cases without computing

Every format above runs auto-graded in AoPS Online courses today. The six problems at right are real homework, shown as students answer them.

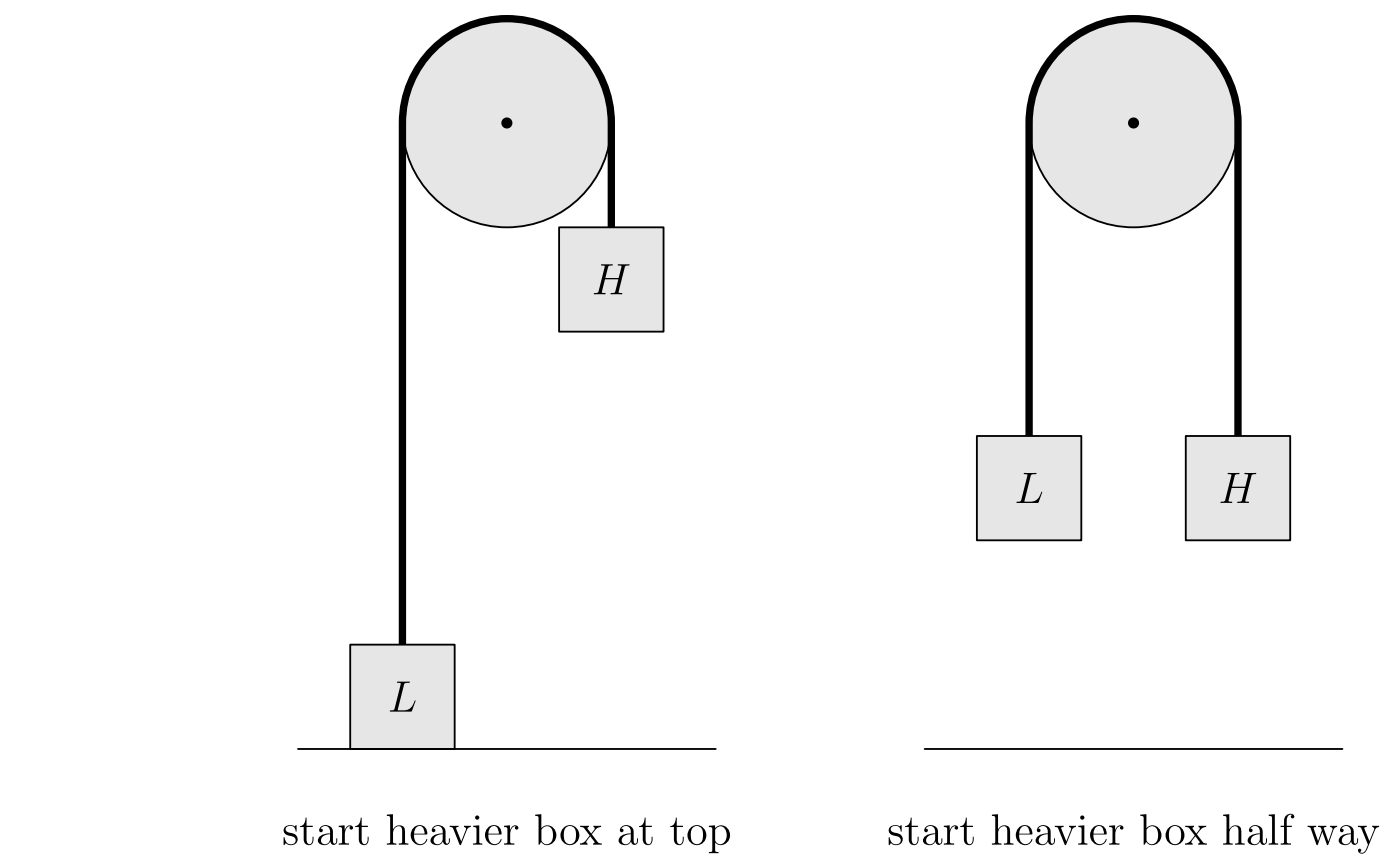
IMPROVE AN EXPERIMENT

Physics 1: Mechanics · Acceleration

Measuring g with an Atwood machine

You're measuring g by releasing an Atwood machine and timing how long the heavier box takes to reach the floor. Which of these changes will reduce the experiment's uncertainty?

- ☐ release the heavier box from halfway down instead of from the top
- ☒ use several different ratios of the two masses instead of a single ratio
- ☒ time multiple trials at each mass ratio instead of one



What students are doing: weighing timing windows (a longer fall is easier to time), model-stressing ranges, and repeated trials — experimental judgment, with no lab report to grade.

WEIGH EVIDENCE

Introduction to Physics · Thought Experiments

Which hypothesis does each observation support?

Galileo claims every object accelerates the same way under gravity. A (fictional) rival, Prof. Impetus, claims denser objects accelerate faster — and that very low-density objects can even accelerate upward. A third scientist, Dr. Vacuum, tries to settle the question with everyday observations, taking each at face value — *never accounting for air drag*. Sort each observation by which hypothesis Dr. Vacuum will conclude it supports.

Galileo Kepler's laws (density never enters) hammer & feather fall together on the Moon a balloon would fall on the Moon	Impetus paper falls faster when crumpled tight helium balloons rise
Both big & small lead balls land together	Neither a skydiver falls faster head-first

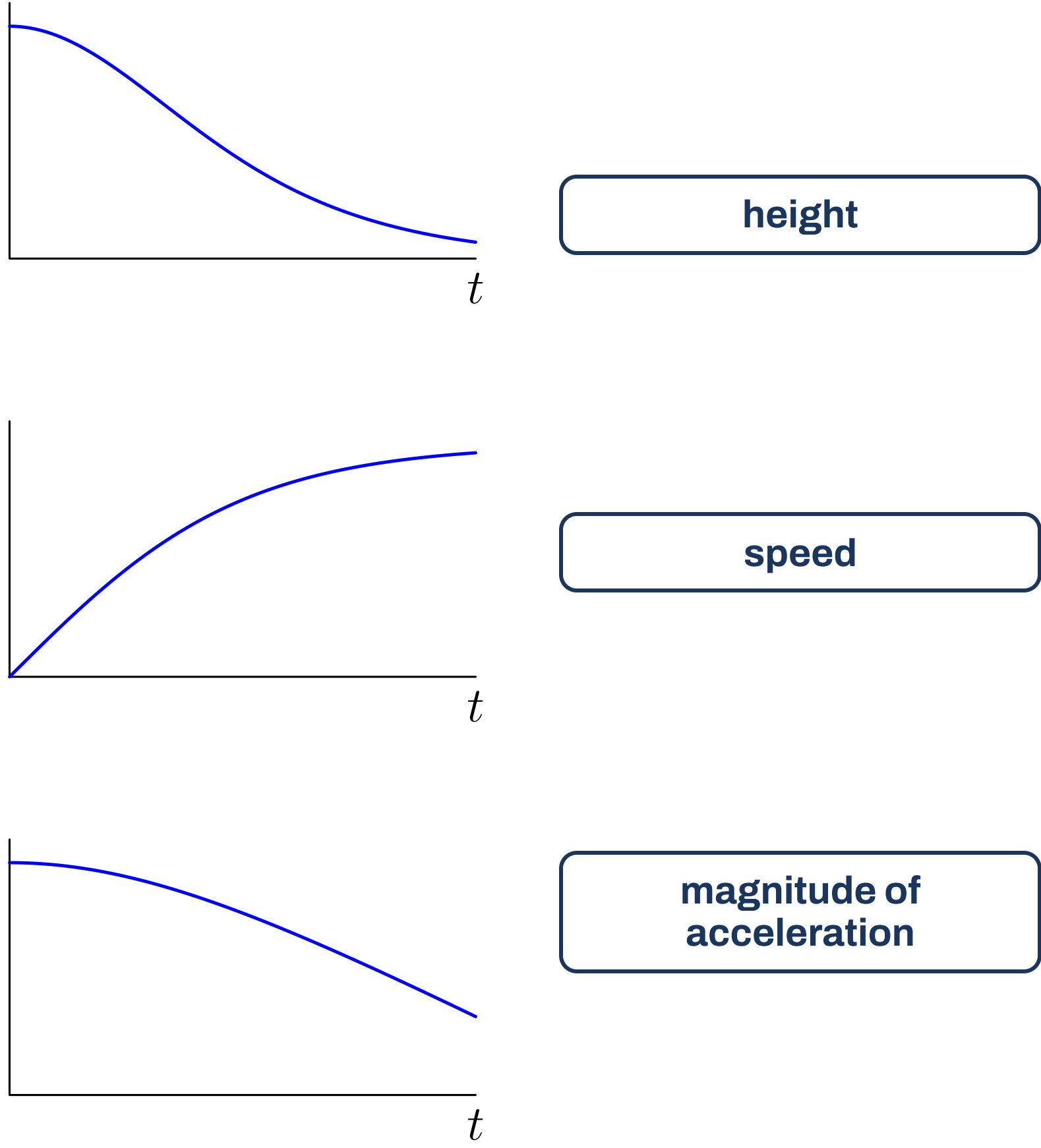
What students are doing: modeling a *flawed* reasoner's inferences — separating what an experiment seems to show from what it shows. Metacognition, graded as a partition.

INTERPRET GRAPHS

Physics 1: Mechanics · Forms of Energy

A peregrine falcon dives

A falcon dives from rest, approaching terminal velocity as air drag builds. Connect each plot to the quantity being plotted.



What students are doing: translating a physical story into qualitative features of curves — speed saturates, acceleration

DIAGNOSE A STUCK SOLVER

Introduction to Physics · Problem Solving

What should each student try next?

Students are stuck on a classic puzzle: a string is tied snugly around Earth's equator; a second string is 6 m longer and circles the equator at a uniform height. How high above the ground is it? Match each student's statement to their most productive next step.

- "I don't understand this problem at all. I don't know where to start." → Restate it: what's unknown? what's given?
- "My answer is 10^{12} m — bigger than Earth itself. Something seems wrong." → Check your methods and calculations
- "I know it's about circumference, but I don't know what to do next." → Solve an easier related problem

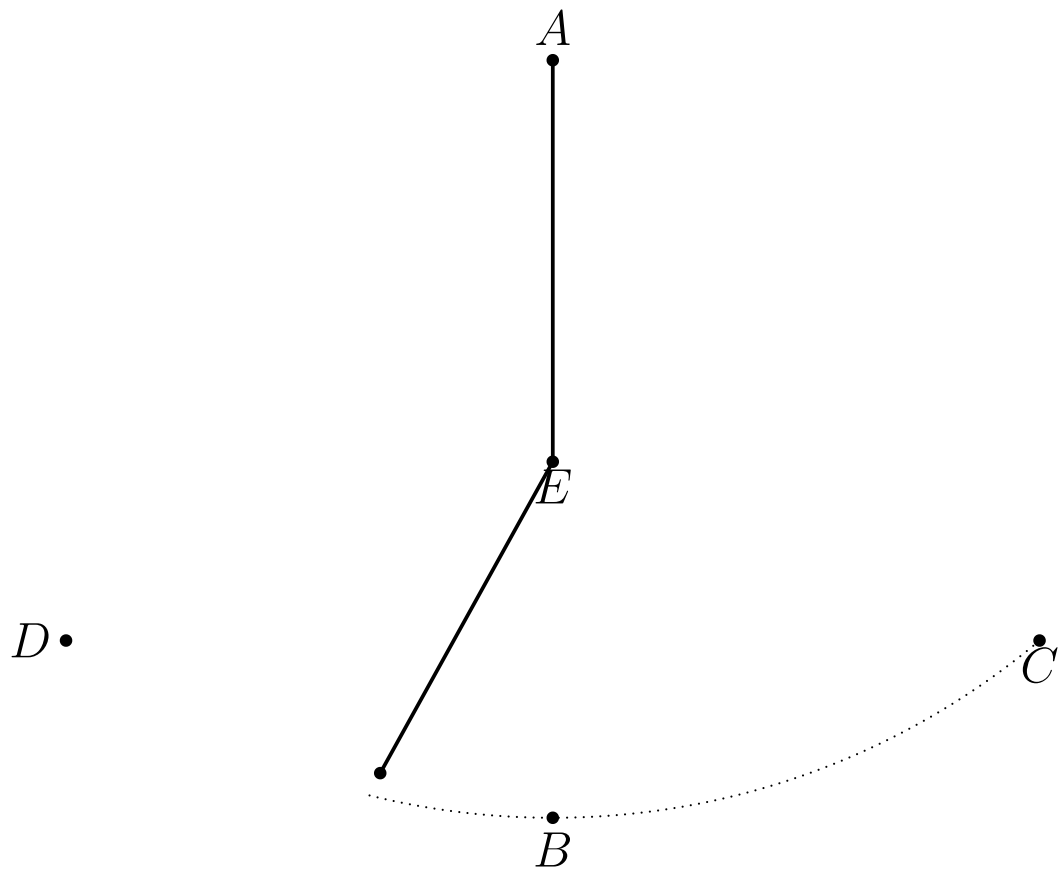
What students are doing: classifying *kinds of stuckness* and prescribing remedies — reasoning about the problem-solving process itself. Pólya, auto-graded.

COMMIT TO A PREDICTION

Introduction to Physics · Conservation Laws

Where does Galileo's pendulum stop?

A pendulum released from C swings through B , where the string catches on a peg at E and the bob swings on around the peg. Click the point in the image where the bob stops and turns around. (The point isn't necessarily labeled.)



What students are doing: turning energy conservation into a *spatial* commitment — same release height, but on the tighter circle around the peg. Clicking D (same height, wrong circle) is a graded near-miss: right law, wrong constraint.

INTERPRET A CALCULATION

Physics 1: Mechanics · Collisions

The calculation disagrees with the cradle

Swing one ball into a Newton's cradle: in the demo, exactly one ball pops out the far side at the incoming speed. But a calculation treating the four resting balls as one block — using only momentum and energy conservation — predicts the incoming ball rebounds while the block slides slowly forward. *Both* outcomes satisfy the conservation laws. Which interpretations are accurate?

- ☐ Momentum or energy must not actually be conserved in the collisions.
- ☒ Elastic collisions + momentum conservation are a good model — but not *enough*; more modeling assumptions are needed.
- ☒ The observed outcome is one solution of the conservation equations; there are many others.
- ☐ The cradle can't be modeled — only experiment can tell us what happens.

What students are doing: deciding what a calculation does and doesn't show — built around a case where theory and demo *disagree*.

What are students doing?

Calculating is one row of this table — not the whole table:

- = apply known equations
- ✓ design a measurement
- ✓ interpret graphs & data
- ✓ build & choose models
- ✓ critique an argument
- ✓ weigh evidence
- ✓ interpret an equation
- ✓ diagnose an error
- ✓ estimate from experience
- ✓ reason about symmetry
- ✓ read & use notation
- ✓ reflect on learning

Every verb above appears in the 28 surveyed problems — each one auto-graded.

Design principles that emerged

- 1** Grade a decision, not a derivation. Bins, matches, and clicks capture judgments directly.
- 2** Put the misconception in the answer key. A wrong bin is a diagnosis ("lighter" at the valley bottom).
- 3** Ask about someone else's thinking. Flawed reasoners and stuck students invite metacognition.
- 4** Let the diagram be the answer. Click-on-image makes a spatial claim gradable.
- 5** Use cases where the model fails. Interpreting a calculation is a gradable skill.
- 6** Structure prevents lucky guessing. Forced pairs and one-to-one matches discipline the format.

Where we're using humans

Human feedback is the precious resource. We spend it where nuance pays most: proofs, writing, open-ended design, and discussion. Broad auto-graded problems don't replace that feedback — they conserve it, by keeping the rest of the homework from defaulting to procedure.

Take it with you



This poster + materials
markeichenlaub.github.io/aapt-sm26-poster



Try these problems, live
four auto-graded demos in your browser

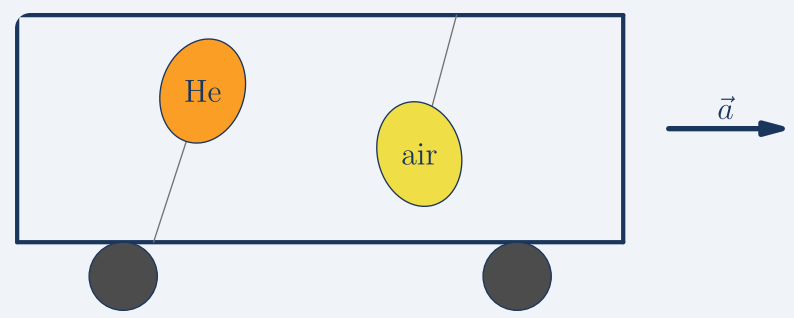


AoPS Online
artofproblemsolving.com/online

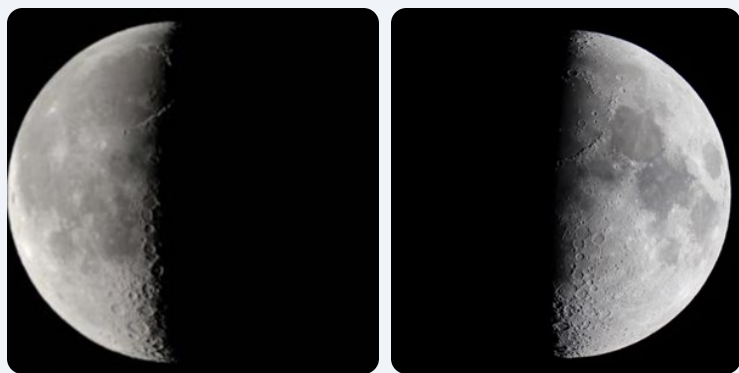
...and more, from a 28-problem survey across four AoPS physics courses



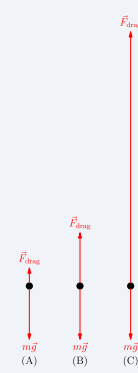
Reason backward from evidence. A tube lamp is either vertical or horizontal. From each object's shadow photo alone, click which. (Middle School Physics)



Predict the counterintuitive. Accelerate the car: the helium balloon leans *forward*. (Mechanics · Fluid Statics)



Reason in 3-D. Which quarter moon rises at sunset, and which at noon? Pure spatial reasoning, graded as a match. (Intro to Physics)



Read a diagram quantitatively. Which free-body diagram is terminal velocity — and which is the chute just opening? Arrow *lengths* carry the physics. (Mechanics)

$$\bar{q} \quad \Delta q \quad q = a \pm b$$

⇕
meanings

Read notation as language. Match decorations on q to what they say. Implicit literacy, made explicit. (Intro to Physics)

$$T = 2\pi\sqrt{\ell/g}$$

plot T^2 vs. ℓ
⇒ slope = $4\pi^2/g$

Linearize real data. Choose axes that straighten pendulum data, then read g off the slope — the working scientist's workflow. (Intro to Physics)